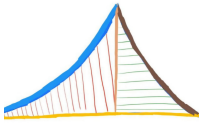
Math for EE



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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE MATHEMATICS 2017 (CLASS 12)

A. Formulae

1.1.1 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{1.1.1.1}$$

The same form represents a plane when $\mathbf{x}$ and $\mathbf{n}$ are 3-dimensional vectors.

1.1.2 The distance between the parallel planes $\mathbf{n}^\top \mathbf{x} = c_1$ and $\mathbf{n}^\top \mathbf{x} = c_2$ is

$$d = \frac{|c_2 - c_1|}{\|\mathbf{n}\|} \tag{1.1.2.1}$$

Solution: The point $\mathbf{x}_1 = \frac{c_1}{\|\mathbf{n}\|^2}\mathbf{n}$ lies on the first plane, from (1.1.1.1). Moving along $\mathbf{n}$, $\mathbf{x}_1 + t\mathbf{n}$ lies on the second plane if

$$\mathbf{n}^\top (\mathbf{x}_1 + t\mathbf{n}) = c_2 \implies c_1 + t\|\mathbf{n}\|^2 = c_2 \implies t = \frac{c_2 - c_1}{\|\mathbf{n}\|^2} \tag{1.1.2.2}$$

The distance is $|t|\,\|\mathbf{n}\|$, yielding (1.1.2.1).

1.1.3 For a square matrix $\mathbf{A}$ of order n,

$$\mathbf{A}\,\mathrm{adj}\,(\mathbf{A}) = \left|\mathbf{A}\right| \mathbf{I} \tag{1.1.3.1}$$

1.1.4 (Trace of a Matrix) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a square matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then,

$$\sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} a_{ii} = \mathrm{tr}\,(\mathbf{A}) \tag{1.1.4.1}$$

1.1.5 The probability mass function (PMF) of the outcome X of a fair die is

$$p_X(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.5.1}$$

The probability of an event E is

$$\Pr(E) = \sum_{n\in E} p_X(n) \tag{1.1.5.2}$$

1.1.6 The events A and B are independent if and only if

$$\Pr(AB) = \Pr(A)\Pr(B) \tag{1.1.6.1}$$

1.1.7 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ if and only if

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = f(x_0) \tag{1.1.7.1}$$

1.1.8 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'\left(x_0^-\right) = \lim_{h\to 0^-} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h\to 0^+} \frac{f(x_0+h) - f(x_0)}{h} = f'\left(x_0^+\right) \tag{1.1.8.1}$$

1.1.9 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$.
Solution:

$$\lim_{x \to x_0} [f(x) - f(x_0)] = \lim_{x \to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] = f'(x_0) \cdot 0 = 0 \tag{1.1.9.1}$$

1.1.10 (Piecewise Boundary Conditions at $x = a$) Consider a function defined by differentiable branches $f_1(x)$ and $f_2(x)$,

$$f(x) = \begin{cases} f_1(x), & x \leq a \\ f_2(x), & x > a \end{cases} \tag{1.1.10.1}$$

For $f(x)$ to be continuous at $x = a$,

$$\lim_{x \to a^-} f_1(x) = \lim_{x \to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{1.1.10.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives must coincide,

$$f_1'(a) = f_2'(a) \tag{1.1.10.3}$$

1.1.11 (Newton-Raphson Method) For a real-valued differentiable function $g(x)$, the root of $g(x) = 0$ is iteratively approximated using the first-order Taylor series expansion about the current estimate x_n, which gives the difference equation

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)} \tag{1.1.11.1}$$

Solution: The first-order Taylor series of $g(x)$ about x_n is

$$g(x) \approx g(x_n) + g'(x_n)(x - x_n) \tag{1.1.11.2}$$

Equating the right side to zero and calling the solution x_{n+1} yields (1.1.11.1). If $x_n \to x^*$, then $x^* = x^* - \frac{g(x^*)}{g'(x^*)}$, so $g(x^*) = 0$, that is, the limit is a root of $g(x)$.

1.1.12 For a quadratic equation $g(x) = ax^2 + bx + c = 0$, we have $g'(x) = 2ax + b$, and the difference equation in (1.1.11.1) becomes

$$x_{n+1} = x_n - \frac{ax_n^2 + bx_n + c}{2ax_n + b} = \frac{ax_n^2 - c}{2ax_n + b} \tag{1.1.12.1}$$

which converges to the root nearest to x_0, provided $2ax_0 + b \neq 0$.

1.1.13 (Rolle's Theorem) If $f(x)$ is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that

$$f'(c) = 0 \tag{1.1.13.1}$$

1.1.14 If $f'(x) > 0$ for all x in an interval, then $f(x)$ is strictly increasing on that interval.

1.1.15 (Trapezoidal Rule) Divide $[a, b]$ into N equal sub-intervals of width $h = \frac{b-a}{N}$, with $x_n = a + nh$. Approximating $f(x)$ by a straight line on each sub-interval, the integral

is computed by the difference equation

$$I_{n+1} = I_n + \frac{h}{2}\left(f\left(x_n\right) + f\left(x_{n+1}\right)\right), \quad I_0 = 0 \tag{1.1.15.1}$$

and $\int_a^b f\left(x\right)\, dx \approx I_N$.

Solution: On $[x_n, x_{n+1}]$, the area under the straight line joining $\left(x_n, f\left(x_n\right)\right)$ and $\left(x_{n+1}, f\left(x_{n+1}\right)\right)$ is a trapezoid,

$$\int_{x_n}^{x_{n+1}} f\left(x\right)\, dx \approx \frac{x_{n+1} - x_n}{2}\left(f\left(x_n\right) + f\left(x_{n+1}\right)\right) = \frac{h}{2}\left(f\left(x_n\right) + f\left(x_{n+1}\right)\right) \tag{1.1.15.2}$$

Adding this area to the integral $I_n = \int_a^{x_n} f\left(x\right)\, dx$ obtained so far yields (1.1.15.1). For a twice differentiable $f\left(x\right)$, the error is

$$\int_a^b f\left(x\right)\, dx - I_N = -\frac{\left(b-a\right)h^2}{12} f''\left(\xi\right), \quad \xi \in \left(a, b\right) \tag{1.1.15.3}$$

so it decreases as h^2.

1.1.16 For a continuous function $f\left(x\right)$,

$$\int_0^a f\left(x\right)\, dx = \int_0^a f\left(a - x\right)\, dx \tag{1.1.16.1}$$

Solution: Substituting $x = a - t$, $dx = -dt$, the limits $x = 0, a$ become $t = a, 0$,

$$\int_0^a f\left(x\right)\, dx = -\int_a^0 f\left(a - t\right)\, dt = \int_0^a f\left(a - t\right)\, dt \tag{1.1.16.2}$$

1.1.17 Trigonometric identities,

$$\tan\left(\pi - x\right) = -\tan x, \quad \sec\left(\pi - x\right) = -\sec x \tag{1.1.17.1}$$

$$\frac{\tan x}{\sec x + \tan x} = \frac{\sin x}{1 + \sin x}, \quad \frac{1}{1 + \sin x} = \sec^2 x - \sec x \tan x \tag{1.1.17.2}$$

Solution: Multiplying the numerator and the denominator by $\cos x$ gives the first identity. For the second,

$$\frac{1}{1 + \sin x} = \frac{1 - \sin x}{1 - \sin^2 x} = \frac{1 - \sin x}{\cos^2 x} \tag{1.1.17.3}$$

which equals $\sec^2 x - \sec x \tan x$.

1.1.18 The derivatives of $\tan x$ and $\sec x$ are

$$\frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}\sec x = \sec x \tan x \tag{1.1.18.1}$$

1.1.19 The absolute value is defined as

$$\left|x - a\right| = \begin{cases} x - a, & x \geq a \\ a - x, & x < a \end{cases} \tag{1.1.19.1}$$

1.1.20 Two vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if

$$\mathbf{u}^{\top}\mathbf{v} = 0 \tag{1.1.20.1}$$

1.1.21 The area of a triangle with two sides $\mathbf{u}$ and $\mathbf{v}$ meeting at a vertex is

$$A = \frac{1}{2}\left\|\mathbf{u}\times\mathbf{v}\right\| \tag{1.1.21.1}$$

where, for $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$,

$$\mathbf{u}\times\mathbf{v} = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix} \tag{1.1.21.2}$$

1.1.22 The mean and the variance of a discrete random variable X with PMF $p_X(n)$ are

$$E(X) = \sum_n n\,p_X(n), \quad \text{Var}(X) = \sum_n n^2\,p_X(n) - (E(X))^2 \tag{1.1.22.1}$$

1.1.23 (Sum of Two Cards Drawn from $\{1,3,5,7\}$)

- Let $X_1, X_2 \in \{1,3,5,7\}$ be independent discrete uniform random variables representing the numbers on two cards drawn with replacement, each having the PMF

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{4}, & n \in \{1,3,5,7\} \\ 0, & \text{otherwise} \end{cases} \tag{1.1.23.1}$$

- Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$,

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n-k \mid X_2 = k)\,p_{X_2}(k) \tag{1.1.23.2}$$

$$= \sum_k p_{X_1}(n-k)\,p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \tag{1.1.23.3}$$

- Substituting (1.1.23.1) into (1.1.23.3) yields

$$p_X(n) = \begin{cases} \frac{1}{16}, & n = 2, 14 \\ \frac{2}{16}, & n = 4, 12 \\ \frac{3}{16}, & n = 6, 10 \\ \frac{4}{16}, & n = 8 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.23.4}$$

- The cards are drawn without replacement, so the case $X_1 = X_2$ is not possible. It contributes $\frac{1}{16}$ to $n = 2, 6, 10, 14$, and $\Pr(X_1 = X_2) = \frac{1}{4}$. Removing it and

normalising by $1-\frac{1}{4}$ gives the PMF

$$p_X(n) = \begin{cases} \frac{1}{6}, & n = 4, 6, 10, 12 \\ \frac{1}{3}, & n = 8 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.23.5}$$

The following code verifies (1.1.23.5) in Fig. 1.1.23.1

codes/PMF.py

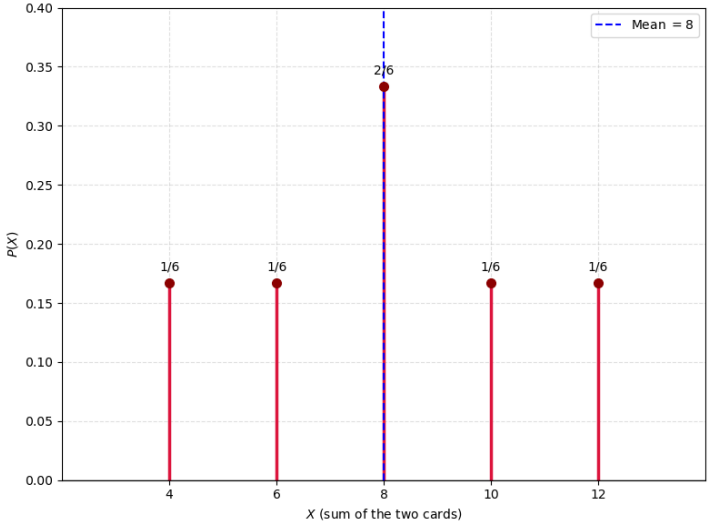


Fig. 1.1.23.1: Probability mass function

1.1.24 (Bayes' Theorem) Let $E_1, E_2, \ldots, E_k$ be mutually exclusive events covering the sample space. For an event A,

$$\Pr(E_i|A) = \frac{\Pr(E_i)\Pr(A|E_i)}{\sum_{j=1}^{k}\Pr(E_j)\Pr(A|E_j)} \tag{1.1.24.1}$$

1.1.25 The value of a determinant does not change if a multiple of one row (or column) is subtracted from another row (or column). A determinant can be expanded along any row or column, using the cofactors of its entries.

1.1.26 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{1.1.26.1}$$

is consistent if and only if

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \tag{1.1.26.2}$$

and the solution is unique if r equals the number of unknowns.

1.1.27 (Least Squares) If $\mathbf{Ax} = \mathbf{b}$ has more equations than unknowns, the least squares solution minimises

$$\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|^2 \implies \mathbf{x} = \left(\mathbf{A}^\top \mathbf{A}\right)^{-1} \mathbf{A}^\top \mathbf{b} \tag{1.1.27.1}$$

provided rank $(\mathbf{A})$ equals the number of unknowns. If the system is consistent, the minimum value is zero and $\mathbf{x}$ is the exact solution. **Solution:** Expanding the cost,

$$f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{Ax} - 2\mathbf{b}^\top \mathbf{Ax} + \mathbf{b}^\top \mathbf{b} \tag{1.1.27.2}$$

Setting the gradient to zero,

$$2\mathbf{A}^\top \mathbf{Ax} - 2\mathbf{A}^\top \mathbf{b} = \mathbf{0} \implies \mathbf{A}^\top \mathbf{Ax} = \mathbf{A}^\top \mathbf{b} \tag{1.1.27.3}$$

The matrix $\mathbf{A}^\top \mathbf{A}$ is invertible when rank $(\mathbf{A})$ equals the number of unknowns, which gives (1.1.27.1).

1.1.28 The matrix of the points $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ is $\mathbf{P} = \begin{pmatrix}\mathbf{A} & \mathbf{B} & \mathbf{C} & \mathbf{D}\end{pmatrix}$, and their direction matrix is

$$\mathbf{M} = \begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} & \mathbf{D}-\mathbf{A}\end{pmatrix} = \mathbf{P}\begin{pmatrix}-1 & -1 & -1\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix} \tag{1.1.28.1}$$

The points are coplanar if and only if

$$\text{rank}\,(\mathbf{M}) < 3 \tag{1.1.28.2}$$

that is, the determinant of $\mathbf{M}$ is zero.

1.1.29 The plane through the points $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ is

$$\mathbf{x} = \mathbf{A} + \lambda(\mathbf{B} - \mathbf{A}) + \mu(\mathbf{C} - \mathbf{A}), \quad \lambda, \mu \in \mathbb{R} \tag{1.1.29.1}$$

The point of intersection of this plane with the line in (1.1.30.1) is obtained by solving the linear system for λ, μ and κ, which has a unique solution if the rank of the coefficient matrix is 3.

1.1.30 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa\mathbf{m} \tag{1.1.30.1}$$

where $\mathbf{m}$ is the direction vector, and $\mathbf{m}^\top \mathbf{n} = 0$ for a normal $\mathbf{n}$ of the line.

1.1.31 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ $(\kappa \in \mathbb{R})$ with a conic section $\mathbf{x}^\top \mathbf{Vx} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \tag{1.1.31.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top(\mathbf{Vh}+\mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top(\mathbf{Vh}+\mathbf{u})\right]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{Vm})}}{\mathbf{m}^\top \mathbf{Vm}}, \quad i = 1, 2 \tag{1.1.31.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

1.1.32 The area enclosed between the curves $y = f(x)$ (upper) and $y = g(x)$ (lower) from $x = a$ to $x = b$ is

$$A = \int_a^b [f(x) - g(x)]\, dx \tag{1.1.32.1}$$

1.1.33 A function $f : X \to Y$ is bijective if it is one-one and onto,

$$f(x_1) = f(x_2) \implies x_1 = x_2, \quad \forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y \tag{1.1.33.1}$$

The inverse of a bijective function is obtained by solving $y = f(x)$ for x.

1.1.34 The function $f(x) = \frac{ax+b}{cx+d}$ corresponds to the matrix $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ acting on $\begin{pmatrix} x \\ 1 \end{pmatrix}$. Composition of such functions corresponds to the matrix product, and the inverse function corresponds to $\mathbf{M}^{-1}$ when $|\mathbf{M}| \neq 0$.

1.1.35 (Inverse by Gauss-Jordan Elimination) The inverse of an $n \times n$ matrix $\mathbf{A}$ exists if and only if rank $(\mathbf{A}) = n$. It is obtained by applying elementary row operations to the augmented matrix until the left half becomes the identity,

$$[\mathbf{A} \mid \mathbf{I}] \xrightarrow{\text{row operations}} \left[\mathbf{I} \mid \mathbf{A}^{-1}\right] \tag{1.1.35.1}$$

and the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. **Solution:** Each row operation is multiplication on the left by an elementary matrix. If $\mathbf{E}$ is the product of the elementary matrices that reduce $\mathbf{A}$ to $\mathbf{I}$, then

$$\mathbf{E}[\mathbf{A} \mid \mathbf{I}] = [\mathbf{E}\mathbf{A} \mid \mathbf{E}] = [\mathbf{I} \mid \mathbf{E}] \tag{1.1.35.2}$$

so $\mathbf{E}\mathbf{A} = \mathbf{I}$ and $\mathbf{E} = \mathbf{A}^{-1}$.

1.1.36 The Laplace transform of a function $g(t)$ is defined as

$$g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s) = \int_0^\infty e^{-st} g(t)\, dt \tag{1.1.36.1}$$

The independent variable can have any name, such as y.

1.1.37 The Laplace transform of the derivative is

$$g'(t) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - g(0) \tag{1.1.37.1}$$

Solution: Applying integration by parts, with $u = e^{-st}$ and $dv = g'(t)\, dt$,

$$\int_0^\infty e^{-st} g'(t)\, dt = \left[e^{-st} g(t)\right]_0^\infty + s \int_0^\infty e^{-st} g(t)\, dt = -g(0) + sG(s) \tag{1.1.37.2}$$

1.1.38 Multiplication by t corresponds to differentiation with respect to s,

$$t\, g(t) \overset{\mathcal{L}}{\longleftrightarrow} -\frac{dG(s)}{ds} \tag{1.1.38.1}$$

Solution: Differentiating (1.1.36.1) with respect to s,

$$\frac{dG(s)}{ds} = \int_0^\infty (-t)\, e^{-st} g(t)\, dt \tag{1.1.38.2}$$

1.1.39 For a non-negative integer n,

$$t^n u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{n!}{s^{n+1}} \tag{1.1.39.1}$$

Solution: From $u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s}$, applying (1.1.38.1) n times gives $(-1)^n \frac{d^n}{ds^n}\frac{1}{s} = \frac{n!}{s^{n+1}}$.

1.1.40 (Shift Property) For a constant a,

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{1.1.40.1}$$

1.1.41 The Laplace transforms of the cosine and sine functions are

$$\cos(at)\, u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2}, \quad \sin(at)\, u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \tag{1.1.41.1}$$

1.1.42 The linear system $\frac{d\mathbf{x}}{dt} = \mathbf{M}\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$ has the Laplace transform

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{M})^{-1}\mathbf{x}_0 \tag{1.1.42.1}$$

Solution: Taking the Laplace transform of each row and using (1.1.37.1),

$$s\mathbf{X}(s) - \mathbf{x}_0 = \mathbf{M}\mathbf{X}(s) \implies (s\mathbf{I} - \mathbf{M})\mathbf{X}(s) = \mathbf{x}_0 \tag{1.1.42.2}$$

1.1.43 For a 2×2 matrix $\mathbf{M}$, the poles of $(s\mathbf{I} - \mathbf{M})^{-1}$ are the eigenvalues of $\mathbf{M}$, and

$$\left|s\mathbf{I} - \mathbf{M}\right| = s^2 - \operatorname{tr}(\mathbf{M})\, s + \left|\mathbf{M}\right| \tag{1.1.43.1}$$

The sum of the eigenvalues is $\operatorname{tr}(\mathbf{M})$ and their product is $\left|\mathbf{M}\right|$.

B. Problems

1.2.1 Find the distance between the planes $2x - y + 2z = 5$ and $5x - 2.5y + 5z = 20$. (CBSE 2017)

Solution: The first plane is, from (1.1.1.1),

$$\begin{pmatrix} 2 & -1 & 2 \end{pmatrix}\mathbf{x} = 5 \tag{1.2.1.1}$$

Dividing the second equation by 2.5,

$$\begin{pmatrix} 5 & -2.5 & 5 \end{pmatrix}\mathbf{x} = 20 \implies \begin{pmatrix} 2 & -1 & 2 \end{pmatrix}\mathbf{x} = 8 \tag{1.2.1.2}$$

The planes are parallel with

$$\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad c_1 = 5, \quad c_2 = 8, \quad \|\mathbf{n}\| = \sqrt{4+1+4} = 3 \tag{1.2.1.3}$$

From (1.1.2.1),

$$d = \frac{|8-5|}{3} = 1 \tag{1.2.1.4}$$

The following code verifies the solution

codes/distance.py

1.2.2 If for any 2×2 square matrix $\mathbf{A}$, $\mathbf{A}\,\mathrm{adj}\,(\mathbf{A}) = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$, then write the value of $\left|\mathbf{A}\right|$. (CBSE 2017)

Solution: From (1.1.3.1),

$$\mathbf{A}\,\mathrm{adj}\,(\mathbf{A}) = \left|\mathbf{A}\right|\mathbf{I} = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} = 8\mathbf{I} \tag{1.2.2.1}$$

Comparing the entries,

$$\left|\mathbf{A}\right| = 8 \tag{1.2.2.2}$$

As a check, taking the trace of both sides using (1.1.4.1),

$$\mathrm{tr}\left(\left|\mathbf{A}\right|\mathbf{I}\right) = 2\left|\mathbf{A}\right| = \mathrm{tr}\,(8\mathbf{I}) = 16 \implies \left|\mathbf{A}\right| = 8 \tag{1.2.2.3}$$

The following code verifies the solution

codes/determinant.py

1.2.3 A die, whose faces are marked 1, 2, 3 in red and 4, 5, 6 in green, is tossed. Let A be the event "number obtained is even" and B be the event "number obtained is red". Find if A and B are independent events. (CBSE 2017)

Solution: The events are

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3\}, \quad AB = \{2\} \tag{1.2.3.1}$$

From (1.1.5.1) and (1.1.5.2),

$$\Pr(A) = \frac{3}{6} = \frac{1}{2}, \quad \Pr(B) = \frac{3}{6} = \frac{1}{2}, \quad \Pr(AB) = \frac{1}{6} \tag{1.2.3.2}$$

From (1.1.6.1),

$$\Pr(A)\Pr(B) = \frac{1}{4} \neq \frac{1}{6} = \Pr(AB) \tag{1.2.3.3}$$

Therefore, A and B are not independent events.
The following code verifies the solution

codes/dice.c

1.2.4 Two tailors, A and B, earn ₹ 300 and ₹ 400 per day respectively. A can stitch 6 shirts and 4 pairs of trousers while B can stitch 10 shirts and 4 pairs of trousers per day. To find how many days should each of them work and if it is desired to produce at least 60 shirts and 32 pairs of trousers at a minimum labour cost, formulate this as an LPP. (CBSE 2017)

Solution: The following code generates Fig. 1.2.4.1

codes/tailor_cost.py

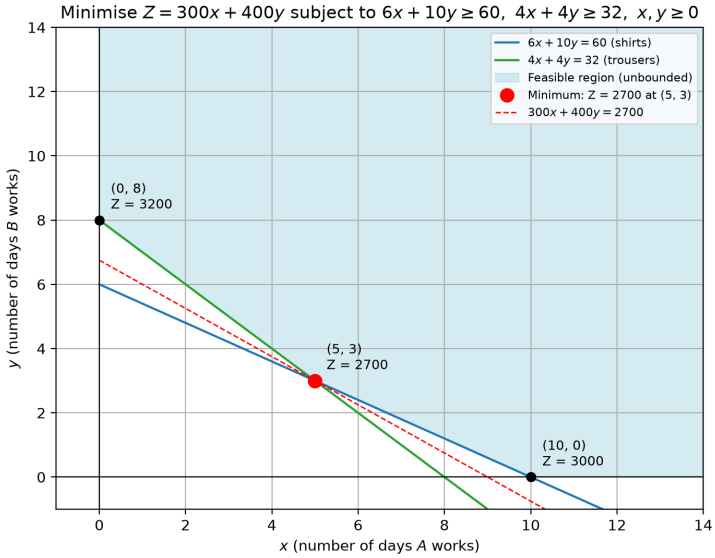


Fig. 1.2.4.1: Feasible region of the LPP

1.2.5 Find the value of c in Rolle's theorem for the function $f(x) = x^3 - 3x$ in $\left[-\sqrt{3}, 0\right]$. (CBSE 2017)

Solution: The function $f(x)$ is a polynomial, so it is differentiable, and hence continuous, from (1.1.9.1). At the end points,

$$f\left(-\sqrt{3}\right) = -3\sqrt{3} + 3\sqrt{3} = 0, \quad f(0) = 0 \tag{1.2.5.1}$$

The conditions of (1.1.13.1) hold, so there exists $c \in \left(-\sqrt{3}, 0\right)$ with $f'(c) = 0$. The derivative is

$$f'(x) = 3x^2 - 3 = 0 \tag{1.2.5.2}$$

which is a quadratic equation. To solve it, apply (1.1.11.1) to $g(x) = f'(x) = 3x^2 - 3$, which is of the form $ax^2 + bx + c$ with $a = 3$, $b = 0$, $c = -3$. From (1.1.12.1), the difference equation is

$$x_{n+1} = \frac{3x_n^2 + 3}{6x_n} = \frac{x_n^2 + 1}{2x_n} \tag{1.2.5.3}$$

Starting with $x_0 = -1.5 \in \left(-\sqrt{3}, 0\right)$,

$$x_1 \approx -1.0833, \quad x_2 \approx -1.0032, \quad x_3 \approx -1.0000 \tag{1.2.5.4}$$

The iterations converge to -1. This can be justified from the difference equation, since

$$x_{n+1}+1=\frac{(x_n+1)^2}{2x_n} \tag{1.2.5.5}$$

For $x_n<0$, the right side is non-positive, so $x_{n+1}\le -1$ after the first step. Also, $|x_n+1|<2|x_n|$, so $|x_{n+1}+1|<|x_n+1|$ and the error keeps decreasing. Hence $x_n\to -1$, which is the same as the root of $3(x-1)(x+1)=0$ lying in $\left(-\sqrt{3},0\right)$, since $-\sqrt{3}\approx -1.732$ and $+1$ is outside the interval. Therefore,

$$c=-1 \tag{1.2.5.6}$$

The tangent at $(c,f(c))=(-1,2)$ is horizontal, and it is parallel to the chord joining the end points, as shown in Fig. 1.2.5.1.
The following code verifies the solution using the difference equation

codes/rolle.c

The following code generates Fig. 1.2.5.1

codes/rolle.py

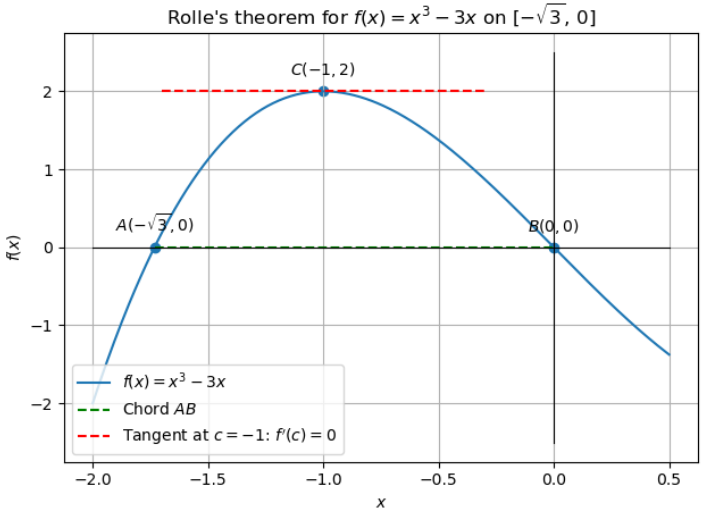


Fig. 1.2.5.1: Graph of $f(x)=x^3-3x$ on $\left[-\sqrt{3},0\right]$ with the horizontal tangent at $c=-1$

1.2.6 Show that the function $f(x)=x^3-3x^2+6x-100$ is increasing on $\mathbb{R}$. (CBSE 2017)

Solution: Differentiating,

$$f'(x) = 3x^2 - 6x + 6 = 3\left[(x-1)^2 + 1\right] \tag{1.2.6.1}$$

Since $(x-1)^2 \geq 0$ for all $x \in \mathbb{R}$,

$$f'(x) \geq 3 > 0 \tag{1.2.6.2}$$

Hence, from (1.1.14), $f(x)$ is strictly increasing on $\mathbb{R}$.
The following code generates Fig. 1.2.6.1

codes/Increasing.py

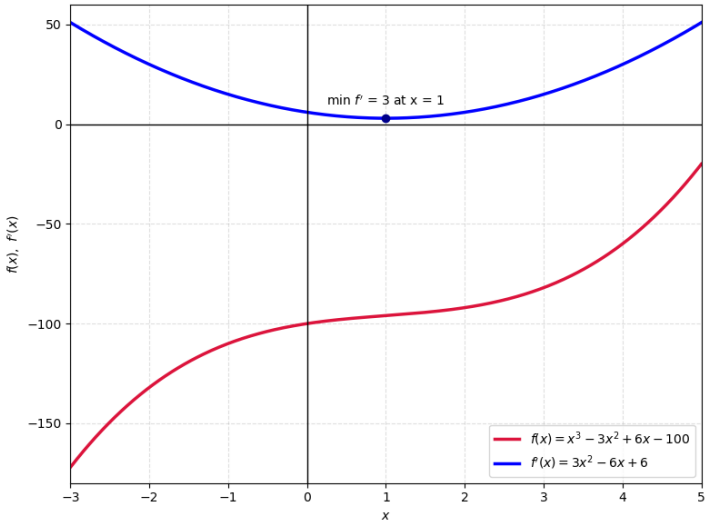


Fig. 1.2.6.1: Increasing function

1.2.7 Evaluate :

$$\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx$$

(CBSE 2017)

Solution: Let

$$I = \int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx \tag{1.2.7.1}$$

From (1.1.16.1) and (1.1.17.1),

$$I = \int_0^{\pi} \frac{(\pi - x)(-\tan x)}{-\sec x - \tan x}\, dx = \int_0^{\pi} \frac{(\pi - x)\tan x}{\sec x + \tan x}\, dx \tag{1.2.7.2}$$

Adding the two expressions of I,

$$2I = \pi \int_0^\pi \frac{\tan x}{\sec x + \tan x}\, dx = \pi \int_0^\pi \frac{\sin x}{1+\sin x}\, dx = \pi \int_0^\pi \left[1 - \frac{1}{1+\sin x}\right] dx \tag{1.2.7.3}$$

from (1.1.17.2). Using (1.1.17.2) and (1.1.18.1),

$$\int_0^\pi \frac{dx}{1+\sin x} = \int_0^\pi \left(\sec^2 x - \sec x \tan x\right) dx = \left[\tan x - \sec x\right]_0^\pi \tag{1.2.7.4}$$

Here $\tan x - \sec x = -\frac{\cos x}{1+\sin x}$ is continuous on $[0, \pi]$, so

$$\left[-\frac{\cos x}{1+\sin x}\right]_0^\pi = 1 - (-1) = 2 \tag{1.2.7.5}$$

Therefore,

$$2I = \pi\,(\pi - 2) \implies I = \frac{\pi\,(\pi-2)}{2} \approx 1.7932 \tag{1.2.7.6}$$

To verify this numerically, from (1.1.17.2) the integrand is

$$f\,(x) = \frac{x \tan x}{\sec x + \tan x} = \frac{x \sin x}{1+\sin x} \tag{1.2.7.7}$$

which is continuous on $[0, \pi]$, even though $\tan x$ is not defined at $x = \frac{\pi}{2}$. Taking N equal sub-intervals of width $h = \frac{\pi}{N}$, with $x_n = nh$, the difference equation of (1.1.15.1) is

$$I_{n+1} = I_n + \frac{h}{2}\left(f\,(x_n) + f\,(x_{n+1})\right), \quad I_0 = 0 \tag{1.2.7.8}$$

and $I \approx I_N$. As N increases, I_N approaches $\frac{\pi(\pi-2)}{2}$, with an error that decreases as h^2.

The following code verifies the solution using the difference equation

codes/Integral_xtanx.c

The following code generates Fig. 1.2.7.1

codes/Integral_xtanx.py

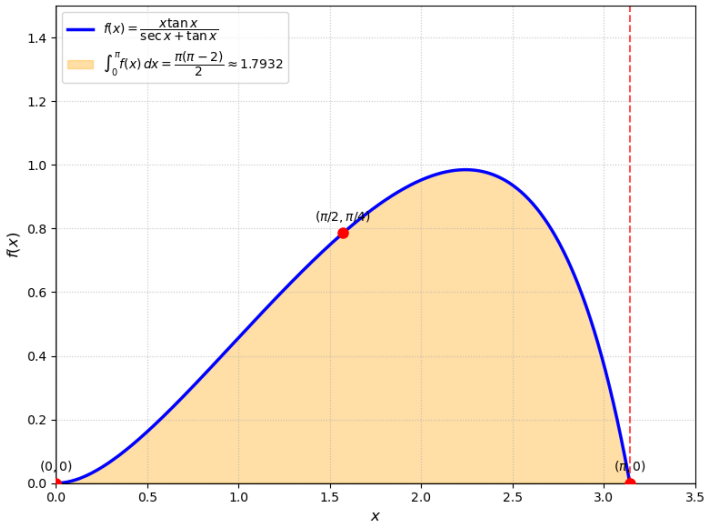


Fig. 1.2.7.1: Integral of $x \tan x$

1.2.8 Evaluate :

$$\int_1^4 \{|x-1| + |x-2| + |x-4|\}\, dx$$

(CBSE 2017)

Solution: From (1.1.19.1), for $1 \le x \le 4$, $|x-1| = x-1$ and $|x-4| = 4-x$, while $|x-2|$ changes at $x = 2$. Thus the integrand is

$$g(x) = \begin{cases} (x-1) + (2-x) + (4-x) = 5-x, & 1 \le x \le 2 \\ (x-1) + (x-2) + (4-x) = x+1, & 2 < x \le 4 \end{cases} \tag{1.2.8.1}$$

At the boundary $x = 2$, from (1.1.10.2),

$$f_1(2) = 5 - 2 = 3 = 2 + 1 = f_2(2) \tag{1.2.8.2}$$

so $g(x)$ is continuous. From (1.1.10.3), $f_1'(2) = -1 \neq 1 = f_2'(2)$, so $g(x)$ is not differentiable at $x = 2$, but it is continuous and hence integrable. Splitting the integral at $x = 2$,

$$\int_1^2 (5-x)\, dx = \left[5x - \frac{x^2}{2}\right]_1^2 = 8 - \frac{9}{2} = \frac{7}{2} \tag{1.2.8.3}$$

$$\int_2^4 (x+1)\, dx = \left[\frac{x^2}{2} + x\right]_2^4 = 12 - 4 = 8 \tag{1.2.8.4}$$

Adding,

$$\int_1^4 \{|x-1|+|x-2|+|x-4|\}\,dx = \frac{7}{2}+8 = \frac{23}{2} \tag{1.2.8.5}$$

The following code generates Fig. 1.2.8.1

codes/Integral_modulus.py

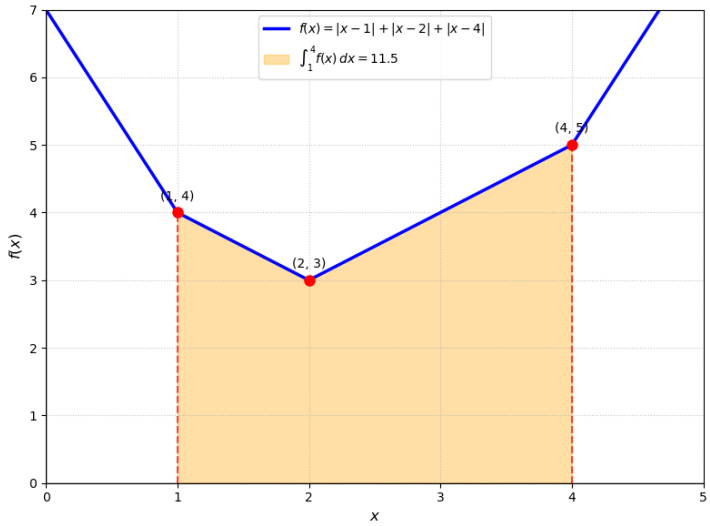


Fig. 1.2.8.1: Integral of the modulus function

1.2.9 Show that the points A, B, C with position vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$ respectively, are the vertices of a right-angled triangle. Hence find the area of the triangle. (CBSE 2017)

Solution: The position vectors are

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 3 \\ -4 \\ -4 \end{pmatrix} \tag{1.2.9.1}$$

The sides are

$$\mathbf{B}-\mathbf{A} = \begin{pmatrix} -1 \\ -2 \\ -6 \end{pmatrix}, \quad \mathbf{C}-\mathbf{B} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{A}-\mathbf{C} = \begin{pmatrix} -1 \\ 3 \\ 5 \end{pmatrix} \tag{1.2.9.2}$$

From (1.1.20.1),

$$(\mathbf{B}-\mathbf{A})^{\top}(\mathbf{C}-\mathbf{B}) = -2+2-6 = -6 \neq 0 \tag{1.2.9.3}$$

$$(\mathbf{C}-\mathbf{B})^{\top}(\mathbf{A}-\mathbf{C}) = -2-3+5 = 0 \tag{1.2.9.4}$$

$$(\mathbf{A}-\mathbf{C})^{\top}(\mathbf{B}-\mathbf{A}) = 1-6-30 = -35 \neq 0 \tag{1.2.9.5}$$

Hence $CB \perp CA$, and the triangle is right angled at C. Also,

$$\|\mathbf{C}-\mathbf{B}\|^2 + \|\mathbf{A}-\mathbf{C}\|^2 = 6+35 = 41 = \|\mathbf{B}-\mathbf{A}\|^2 \tag{1.2.9.6}$$

From (1.1.21.1),

$$(\mathbf{C}-\mathbf{B}) \times (\mathbf{A}-\mathbf{C}) = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} -8 \\ -11 \\ 5 \end{pmatrix} \tag{1.2.9.7}$$

so the area is

$$A = \frac{1}{2}\left\| \begin{pmatrix} -8 \\ -11 \\ 5 \end{pmatrix} \right\| = \frac{1}{2}\sqrt{64+121+25} = \frac{\sqrt{210}}{2} \text{ sq. units} \tag{1.2.9.8}$$

The following code verifies the solution

codes/triangle.py

1.2.10 There are 4 cards numbered 1, 3, 5 and 7, one number on one card. Two cards are drawn at random without replacement. Let X denote the sum of the numbers on the two drawn cards. Find the mean and variance of X. (CBSE 2017)

Solution: Let X_1 and X_2 represent the numbers on the two drawn cards, and let $X = X_1 + X_2$ denote their sum. From the PMF derived in (1.1.23.5),

$$p_X(4) = p_X(6) = p_X(10) = p_X(12) = \frac{1}{6}, \quad p_X(8) = \frac{1}{3} \tag{1.2.10.1}$$

From (1.1.22.1), the mean is

$$E(X) = \frac{4+6+2(8)+10+12}{6} = 8 \tag{1.2.10.2}$$

and

$$\sum_n n^2 p_X(n) = \frac{16+36+2(64)+100+144}{6} = \frac{212}{3} \tag{1.2.10.3}$$

Therefore, the variance is

$$\text{Var}(X) = \frac{212}{3} - 64 = \frac{20}{3} \tag{1.2.10.4}$$

1.2.11 Of the students in a school, it is known that 30% have 100% attendance and 70% students are irregular. Previous year results report that 70% of all students who have 100% attendance attain A grade and 10% irregular students attain A grade in their annual examination. At the end of the year, one student is chosen at random from

the school and he was found to have an A grade. What is the probability that the student has 100% attendance ? Is regularity required only in school ? Justify your answer. (CBSE 2017)

Solution: Let E_1 be the event that the student has 100% attendance, E_2 be the event that the student is irregular and A be the event that the student has an A grade. Then

$$\Pr(E_1) = \frac{3}{10}, \quad \Pr(E_2) = \frac{7}{10}, \quad \Pr(A|E_1) = \frac{7}{10}, \quad \Pr(A|E_2) = \frac{1}{10} \tag{1.2.11.1}$$

From (1.1.24.1),

$$\Pr(E_1|A) = \frac{\frac{3}{10}\cdot\frac{7}{10}}{\frac{3}{10}\cdot\frac{7}{10}+\frac{7}{10}\cdot\frac{1}{10}} = \frac{21}{21+7} = \frac{3}{4} \tag{1.2.11.2}$$

Regularity is not required only in school. It is needed in every area of life, such as studies, work, sports and health habits, because steady and regular effort gives better results. This is seen in the data above, where 70% of the regular students get an A grade against 10% of the irregular students. The following code verifies the solution

codes/bayes.c

1.2.12 Find matrix $\mathbf{A}$ such that

$$\begin{pmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{pmatrix} \mathbf{A} = \begin{pmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{pmatrix}$$

(CBSE 2017)

Solution: Let the equation be $\mathbf{BA} = \mathbf{C}$, where $\mathbf{B}$ is 3×2 and $\mathbf{A}$ is 2×2. The columns of $\mathbf{A}$ satisfy $\mathbf{Ba}_j = \mathbf{c}_j$ for $j = 1, 2$, which is an overdetermined system. From (1.1.27.1),

$$\mathbf{A} = \left(\mathbf{B}^\top\mathbf{B}\right)^{-1}\mathbf{B}^\top\mathbf{C} \tag{1.2.12.1}$$

where

$$\mathbf{B}^\top\mathbf{B} = \begin{pmatrix} 2 & 1 & -3 \\ -1 & 0 & 4 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} 14 & -14 \\ -14 & 17 \end{pmatrix} \tag{1.2.12.2}$$

with $\left|\mathbf{B}^\top\mathbf{B}\right| = 238 - 196 = 42 \neq 0$, so

$$\left(\mathbf{B}^\top\mathbf{B}\right)^{-1} = \frac{1}{42}\begin{pmatrix} 17 & 14 \\ 14 & 14 \end{pmatrix} \tag{1.2.12.3}$$

Also,

$$\mathbf{B}^\top\mathbf{C} = \begin{pmatrix} 2 & 1 & -3 \\ -1 & 0 & 4 \end{pmatrix}\begin{pmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{pmatrix} = \begin{pmatrix} -28 & -84 \\ 37 & 96 \end{pmatrix} \tag{1.2.12.4}$$

Therefore,

$$\mathbf{A} = \frac{1}{42}\begin{pmatrix}17 & 14\\ 14 & 14\end{pmatrix}\begin{pmatrix}-28 & -84\\ 37 & 96\end{pmatrix} = \frac{1}{42}\begin{pmatrix}42 & -84\\ 126 & 168\end{pmatrix} = \begin{pmatrix}1 & -2\\ 3 & 4\end{pmatrix} \quad (1.2.12.5)$$

As a check, $\mathbf{BA} = \mathbf{C}$, so the residual $\|\mathbf{B}\mathbf{a}_j - \mathbf{c}_j\|$ is zero for both columns and the least squares solution is the exact solution.
The following code verifies the solution

codes/matrix.py

1.2.13 Solve the following linear programming problem graphically :
Maximise $Z = 34x + 45y$ under the following constraints

$$x + y \leq 300, \quad 2x + 3y \leq 70, \quad x \geq 0, \ y \geq 0$$

(CBSE 2017)

Solution: The following code generates Fig. 1.2.13.1

codes/LPP.py

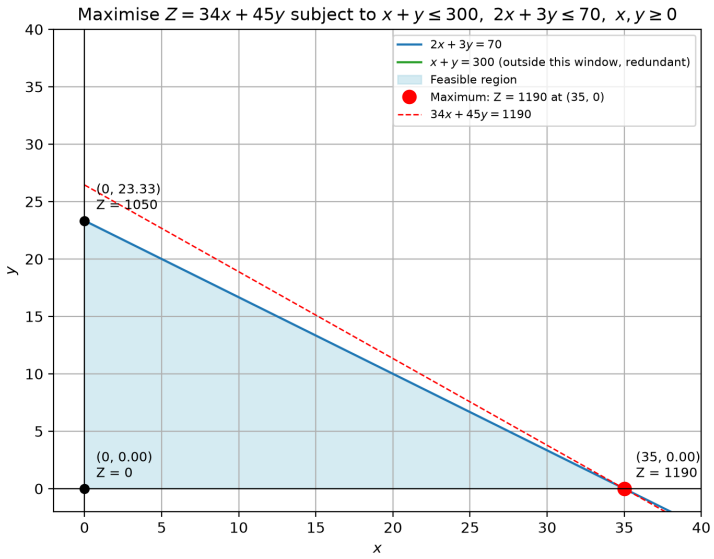


Fig. 1.2.13.1: Linear programming problem

1.2.14 Find the value of x such that the points $A\,(3, 2, 1)$, $B\,(4, x, 5)$, $C\,(4, 2, -2)$ and $D\,(6, 5, -1)$ are coplanar. (CBSE 2017)

Solution: The matrix of the points is

$$\mathbf{P} = \begin{pmatrix} \mathbf{A} & \mathbf{B} & \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} 3 & 4 & 4 & 6 \\ 2 & x & 2 & 5 \\ 1 & 5 & -2 & -1 \end{pmatrix} \tag{1.2.14.1}$$

From (1.1.28.1), the direction matrix is

$$\mathbf{M} = \begin{pmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} & \mathbf{D}-\mathbf{A} \end{pmatrix} = \mathbf{P}\begin{pmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 3 \\ x-2 & 0 & 3 \\ 4 & -3 & -2 \end{pmatrix} \tag{1.2.14.2}$$

The points are coplanar if rank $(\mathbf{M}) < 3$, that is, $\left|\mathbf{M}\right| = 0$. Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - 3C_1$, from (1.1.25),

$$\left|\mathbf{M}\right| = \begin{vmatrix} 1 & 0 & 0 \\ x-2 & -(x-2) & 9-3x \\ 4 & -7 & -14 \end{vmatrix} = 14\,(x-2) + 7\,(9-3x) = 35 - 7x \tag{1.2.14.3}$$

Equating to zero,

$$35 - 7x = 0 \implies x = 5 \tag{1.2.14.4}$$

As a check, for $x = 5$ the columns of $\mathbf{M}$ satisfy $\begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ -2 \end{pmatrix}$, so the rank is less than 3.
The following code verifies the solution

codes/coplanar.py

1.2.15 Find the general solution of the differential equation

$$y\,dx - \left(x + 2y^2\right)\,dy = 0$$

(CBSE 2017)

Solution: Rewriting the equation and multiplying by $y > 0$,

$$\frac{dx}{dy} - \frac{x}{y} = 2y \implies y\frac{dx}{dy} - x = 2y^2 \tag{1.2.15.1}$$

Let $x\,(y) \overset{\mathcal{L}}{\longleftrightarrow} X\,(s)$. From (1.1.38.1) and (1.1.37.1),

$$y\frac{dx}{dy} \overset{\mathcal{L}}{\longleftrightarrow} -\frac{d}{ds}\left[sX\,(s) - x\,(0)\right] = -X\,(s) - sX'\,(s) \tag{1.2.15.2}$$

From (1.1.39.1), $2y^2 \overset{\mathcal{L}}{\longleftrightarrow} \frac{4}{s^3}$. Taking the Laplace transform of (1.2.15.1),

$$-X - sX' - X = \frac{4}{s^3} \tag{1.2.15.3}$$

$$\implies X' + \frac{2}{s}X = -\frac{4}{s^4} \tag{1.2.15.4}$$

Multiplying by s^2,

$$\frac{d}{ds}\left(s^2 X\right) = -\frac{4}{s^2} \tag{1.2.15.5}$$

$$\implies s^2 X = \frac{4}{s} + C \tag{1.2.15.6}$$

$$\implies X\left(s\right) = \frac{4}{s^3} + \frac{C}{s^2} \tag{1.2.15.7}$$

Taking the inverse Laplace transform using (1.1.39.1),

$$x\left(y\right) = 2y^2 + Cy \tag{1.2.15.8}$$

Note that $x\left(0\right)$ cancelled out, so C remains arbitrary. As a check, $\frac{dx}{dy} = 4y + C$ and $\frac{x}{y} + 2y = 2y + C + 2y = 4y + C$. Therefore, the general solution is

$$x = 2y^2 + Cy \tag{1.2.15.9}$$

1.2.16 Find the coordinates of the point where the line through the points $(3, -4, -5)$ and $(2, -3, 1)$, crosses the plane determined by the points $(1, 2, 3)$, $(4, 2, -3)$ and $(0, 4, 3)$. (CBSE 2017)

Solution: Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ be the points of the plane and $\mathbf{D}$, $\mathbf{E}$ be the points of the line. The matrix of the points is

$$\mathbf{P} = \begin{pmatrix}\mathbf{A} & \mathbf{B} & \mathbf{C} & \mathbf{D} & \mathbf{E}\end{pmatrix} = \begin{pmatrix} 1 & 4 & 0 & 3 & 2 \\ 2 & 2 & 4 & -4 & -3 \\ 3 & -3 & 3 & -5 & 1 \end{pmatrix} \tag{1.2.16.1}$$

From (1.1.29.1) and (1.1.30.1), the point of intersection satisfies

$$\mathbf{A} + \lambda\left(\mathbf{B} - \mathbf{A}\right) + \mu\left(\mathbf{C} - \mathbf{A}\right) = \mathbf{D} + \kappa\left(\mathbf{E} - \mathbf{D}\right) \tag{1.2.16.2}$$

which can be written as

$$\begin{pmatrix}\mathbf{B} - \mathbf{A} & \mathbf{C} - \mathbf{A} & \mathbf{D} - \mathbf{E}\end{pmatrix}\begin{pmatrix}\lambda \\ \mu \\ \kappa\end{pmatrix} = \mathbf{D} - \mathbf{A} \tag{1.2.16.3}$$

The direction matrix

$$\mathbf{M} = \begin{pmatrix}\mathbf{B} - \mathbf{A} & \mathbf{C} - \mathbf{A} & \mathbf{D} - \mathbf{E} & \mathbf{D} - \mathbf{A}\end{pmatrix} \tag{1.2.16.4}$$

is obtained from $\mathbf{P}$ as

$$\mathbf{M} = \mathbf{P}\begin{pmatrix} -1 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ -6 & 0 & -6 & -8 \end{pmatrix} \tag{1.2.16.5}$$

Row reducing the augmented matrix $\mathbf{M}$,

$$\begin{aligned} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ -6 & 0 & -6 & -8 \end{pmatrix} &\xrightarrow{R_3 \to R_3 + 2R_1} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ 0 & -2 & -4 & -4 \end{pmatrix} \\ &\xrightarrow{R_3 \to R_3 + R_2} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ 0 & 0 & -5 & -10 \end{pmatrix} \\ &\xrightarrow{R_3 \to -\frac{1}{5}R_3} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 + R_3} \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 2 & 0 & -4 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow{R_2 \to \frac{1}{2}R_2} \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow{R_1 \to R_1 + R_2} \begin{pmatrix} 3 & 0 & 0 & -2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow{R_1 \to \frac{1}{3}R_1} \begin{pmatrix} 1 & 0 & 0 & -\frac{2}{3} \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \end{aligned} \tag{1.2.16.6}$$

Here the rank of the coefficient matrix and the augmented matrix is 3, which is the number of unknowns. From (1.1.26.2), the solution is unique,

$$\lambda = -\frac{2}{3}, \quad \mu = -2, \quad \kappa = 2 \tag{1.2.16.7}$$

Therefore, the point of intersection is

$$\mathbf{x} = \mathbf{D} + \kappa\left(\mathbf{E} - \mathbf{D}\right) = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix} + 2\begin{pmatrix} -1 \\ 1 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 7 \end{pmatrix} \tag{1.2.16.8}$$

As a check,

$$\mathbf{A}+\lambda\left(\mathbf{B}-\mathbf{A}\right)+\mu\left(\mathbf{C}-\mathbf{A}\right)=\begin{pmatrix}1\\2\\3\end{pmatrix}-\frac{2}{3}\begin{pmatrix}3\\0\\-6\end{pmatrix}-2\begin{pmatrix}-1\\2\\0\end{pmatrix}=\begin{pmatrix}1\\-2\\7\end{pmatrix} \tag{1.2.16.9}$$

The following code verifies the solution

codes/lineplane.py

1.2.17 Find the particular solution of the differential equation $(x-y)\frac{dy}{dx}=(x+2y)$, given that $y=0$ when $x=1$. (CBSE 2017)

Solution: The equation contains the product $y\frac{dy}{dx}$, so it cannot be transformed directly. Introducing a parameter t such that

$$\frac{dx}{dt}=x-y,\quad \frac{dy}{dt}=x+2y \tag{1.2.17.1}$$

gives $\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{x+2y}{x-y}$, which is the given equation. In matrix form,

$$\frac{d}{dt}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}1&-1\\1&2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix},\quad \begin{pmatrix}x\\y\end{pmatrix}_{t=0}=\begin{pmatrix}1\\0\end{pmatrix} \tag{1.2.17.2}$$

Let $\mathbf{M}=\begin{pmatrix}1&-1\\1&2\end{pmatrix}$. Then $\text{tr}\left(\mathbf{M}\right)=3$ and $\left|\mathbf{M}\right|=3$. From (1.1.43.1),

$$\left|s\mathbf{I}-\mathbf{M}\right|=s^2-3s+3=\left(s-\frac{3}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2 \tag{1.2.17.3}$$

so the eigenvalues are $\frac{3\pm j\sqrt{3}}{2}$. From (1.1.42.1),

$$\begin{pmatrix}X\left(s\right)\\Y\left(s\right)\end{pmatrix}=\begin{pmatrix}s-1&1\\-1&s-2\end{pmatrix}^{-1}\begin{pmatrix}1\\0\end{pmatrix}=\frac{1}{s^2-3s+3}\begin{pmatrix}s-2\\1\end{pmatrix} \tag{1.2.17.4}$$

Writing $s-2=\left(s-\frac{3}{2}\right)-\frac{1}{2}$,

$$X\left(s\right)=\frac{s-\frac{3}{2}}{\left(s-\frac{3}{2}\right)^2+\frac{3}{4}}-\frac{1}{\sqrt{3}}\cdot\frac{\frac{\sqrt{3}}{2}}{\left(s-\frac{3}{2}\right)^2+\frac{3}{4}} \tag{1.2.17.5}$$

$$Y\left(s\right)=\frac{2}{\sqrt{3}}\cdot\frac{\frac{\sqrt{3}}{2}}{\left(s-\frac{3}{2}\right)^2+\frac{3}{4}} \tag{1.2.17.6}$$

Taking the inverse Laplace transform using (1.1.40.1) and (1.1.41.1), with $\theta=\frac{\sqrt{3}}{2}t$,

$$x\left(t\right)=e^{\frac{3t}{2}}\left(\cos\theta-\frac{1}{\sqrt{3}}\sin\theta\right) \tag{1.2.17.7}$$

$$y\left(t\right)=\frac{2}{\sqrt{3}}e^{\frac{3t}{2}}\sin\theta \tag{1.2.17.8}$$

To eliminate t,

$$x+2y=2e^{\frac{3t}{2}}\sin\left(\theta+\frac{\pi}{6}\right),\quad \sqrt{3}x=2e^{\frac{3t}{2}}\cos\left(\theta+\frac{\pi}{6}\right) \tag{1.2.17.9}$$

$$\implies \frac{x+2y}{\sqrt{3}x}=\tan\left(\theta+\frac{\pi}{6}\right) \tag{1.2.17.10}$$

$$\implies \tan^{-1}\left(\frac{x+2y}{\sqrt{3}x}\right)=\frac{\sqrt{3}}{2}t+\frac{\pi}{6} \tag{1.2.17.11}$$

Also, since $x+\frac{y}{2}=e^{\frac{3t}{2}}\cos\theta$,

$$x^2+xy+y^2=\left(x+\frac{y}{2}\right)^2+\frac{3}{4}y^2=e^{3t}\left(\cos^2\theta+\sin^2\theta\right)=e^{3t} \tag{1.2.17.12}$$

$$\implies \frac{1}{2}\ln\left(x^2+xy+y^2\right)=\frac{3t}{2} \tag{1.2.17.13}$$

Eliminating t from the last two results, the particular solution is

$$\sqrt{3}\tan^{-1}\left(\frac{x+2y}{\sqrt{3}x}\right)-\frac{1}{2}\ln\left(x^2+xy+y^2\right)=\frac{\pi}{2\sqrt{3}} \tag{1.2.17.14}$$

which holds on the branch through $(1,0)$. As a check, at $(1,0)$ the left side is $\sqrt{3}\cdot\frac{\pi}{6}-0=\frac{\pi}{2\sqrt{3}}$.
The following code generates Fig. 1.2.17.1

codes/euler.py

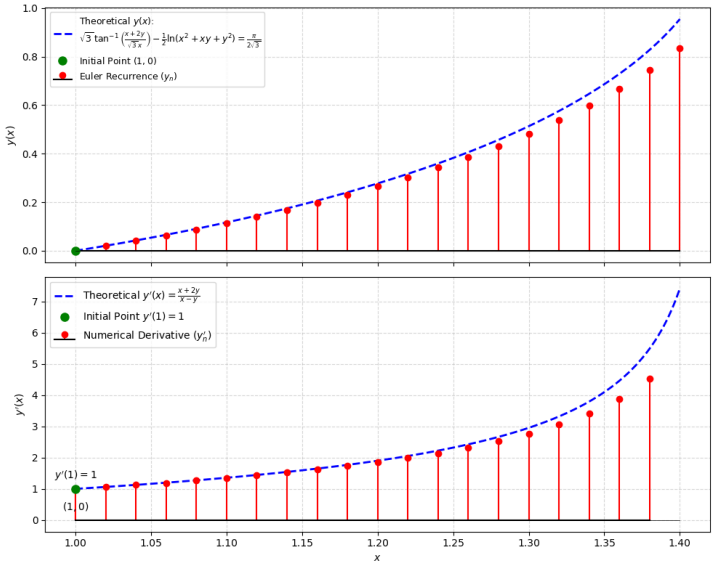


Fig. 1.2.17.1: Euler's method

1.2.18 Find the area enclosed between the parabola $4y = 3x^2$ and the straight line $3x - 2y + 12 = 0$. (CBSE 2017)

Solution: The parabola $3x^2 - 4y = 0$ is a conic with

$$\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad f = 0 \tag{1.2.18.1}$$

The line $3x - 2y + 12 = 0$ is $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ with

$$\mathbf{h} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \tag{1.2.18.2}$$

Then

$$\mathbf{Vh} + \mathbf{u} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad \mathbf{m}^\top (\mathbf{Vh} + \mathbf{u}) = -6, \quad g(\mathbf{h}) = -24, \quad \mathbf{m}^\top \mathbf{Vm} = 12 \tag{1.2.18.3}$$

From (1.1.31.1),

$$\kappa = \frac{6 \pm \sqrt{36 + 288}}{12} = \frac{6 \pm 18}{12} = 2 \text{ or } -1 \tag{1.2.18.4}$$

The points of intersection are

$$\begin{pmatrix} 0 \\ 6 \end{pmatrix} + 2\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 12 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \tag{1.2.18.5}$$

The line is above the parabola between $x = -2$ and $x = 4$. From (1.1.32.1),

$$A = \int_{-2}^{4} \left[\frac{3x+12}{2} - \frac{3x^2}{4}\right] dx = \left[\frac{3x^2}{4} + 6x - \frac{x^3}{4}\right]_{-2}^{4} \tag{1.2.18.6}$$

$$= 20 - (-7) = 27 \text{ sq. units} \tag{1.2.18.7}$$

The following code generates Fig. 1.2.18.1

codes/parabola_line.py

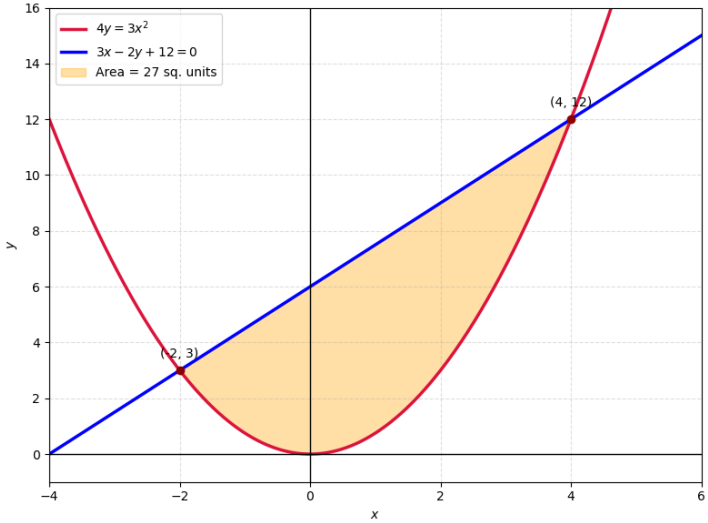


Fig. 1.2.18.1: Parabola and line

1.2.19 Consider $f : \mathbb{R} - \left\{-\frac{4}{3}\right\} \to \mathbb{R} - \left\{\frac{4}{3}\right\}$ given by $f(x) = \frac{4x+3}{3x+4}$. Show that f is bijective. Find the inverse of f and hence find $f^{-1}(0)$ and x such that $f^{-1}(x) = 2$. (CBSE 2017)

Solution: From (1.1.33.1), for one-one,

$$f(x_1) = f(x_2) \implies (4x_1+3)(3x_2+4) = (4x_2+3)(3x_1+4) \tag{1.2.19.1}$$

$$\implies 7x_1 = 7x_2 \implies x_1 = x_2 \tag{1.2.19.2}$$

For onto, let $y = \frac{4x+3}{3x+4}$ with $y \neq \frac{4}{3}$. Solving for x,

$$x = \frac{3-4y}{3y-4} \tag{1.2.19.3}$$

which is defined for every $y \neq \frac{4}{3}$, and $x \neq -\frac{4}{3}$, since $x = -\frac{4}{3}$ would give $-9 = -16$.

So f is onto and hence bijective. The inverse is

$$f^{-1}(x) = \frac{3-4x}{3x-4} \tag{1.2.19.4}$$

From (1.1.34), $\mathbf{M} = \begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$ with $\left|\mathbf{M}\right| = 7 \neq 0$, and $\mathbf{M}^{-1} = \frac{1}{7}\begin{pmatrix} 4 & -3 \\ -3 & 4 \end{pmatrix}$ gives the same inverse $\frac{4x-3}{-3x+4}$. Then

$$f^{-1}(0) = \frac{3}{-4} = -\frac{3}{4} \tag{1.2.19.5}$$

and

$$f^{-1}(x) = 2 \implies 3-4x = 6x-8 \implies x = \frac{11}{10} \tag{1.2.19.6}$$

1.2.20 Let $A = \mathbb{Q} \times \mathbb{Q}$ and let $*$ be a binary operation on A defined by $(a,b) * (c,d) = (ac, b+ad)$ for $(a,b), (c,d) \in A$. Determine, whether $*$ is commutative and associative. Then, with respect to $*$ on A

a) Find the identity element in A.
b) Find the invertible elements of A.

(CBSE 2017)

Solution: Writing (a,b) as the matrix $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$, the product

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ac & ad+b \\ 0 & 1 \end{pmatrix} \tag{1.2.20.1}$$

is exactly $(a,b) * (c,d)$.

a) **Commutative:** $(a,b) * (c,d) = (ac, b+ad)$ whereas $(c,d) * (a,b) = (ac, d+cb)$. For $(1,2) * (3,4) = (3,6)$ and $(3,4) * (1,2) = (3,10)$. So $*$ is not commutative.
b) **Associative:**

$$[(a,b) * (c,d)] * (e,f) = (ac, b+ad) * (e,f) = (ace, b+ad+acf) \tag{1.2.20.2}$$
$$(a,b) * [(c,d) * (e,f)] = (a,b) * (ce, d+cf) = (ace, b+ad+acf) \tag{1.2.20.3}$$

The two are equal, so $*$ is associative.
c) **Identity:** Let (e_1, e_2) be the identity. Then $(a,b) * (e_1, e_2) = (ae_1, b+ae_2) = (a,b)$ for all a, b, giving $e_1 = 1$, $e_2 = 0$. The identity element is $(1,0)$, which also satisfies $(1,0) * (a,b) = (a,b)$.
d) **Invertible elements:** For (c,d) to be the inverse of (a,b),

$$(ac, b+ad) = (1,0) \implies c = \frac{1}{a}, \quad d = -\frac{b}{a} \tag{1.2.20.4}$$

which requires $a \neq 0$. So the invertible elements are (a,b) with $a \neq 0$, and the inverse is $\left(\frac{1}{a}, -\frac{b}{a}\right)$. In matrix form, this is $\left|\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\right| = a \neq 0$.

1.2.21 If $\mathbf{A} = \begin{pmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix}$, find $\mathbf{A}^{-1}$. Hence using $\mathbf{A}^{-1}$ solve the system of equations $2x - 3y + 5z = 11$, $3x + 2y - 4z = -5$, $x + y - 2z = -3$. (CBSE 2017)

Solution: Row reducing the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$,

$$[\mathbf{A} \mid \mathbf{I}] = \begin{pmatrix} 2 & -3 & 5 & 1 & 0 & 0 \\ 3 & 2 & -4 & 0 & 1 & 0 \\ 1 & 1 & -2 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 3 & 2 & -4 & 0 & 1 & 0 \\ 2 & -3 & 5 & 1 & 0 & 0 \end{pmatrix} \tag{1.2.21.1}$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 3R_1} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 0 & -1 & 2 & 0 & 1 & -3 \\ 0 & -5 & 9 & 1 & 0 & -2 \end{pmatrix} \xrightarrow[R_2 \to -R_2]{R_3 \to R_3 - 5R_2'} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 0 & 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & -1 & 1 & -5 & 13 \end{pmatrix} \tag{1.2.21.2}$$

$$\xrightarrow{R_3 \to -R_3} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 0 & 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & -1 & 5 & -13 \end{pmatrix} \xrightarrow[R_1 \to R_1 + 2R_3]{R_2 \to R_2 + 2R_3} \begin{pmatrix} 1 & 1 & 0 & -2 & 10 & -25 \\ 0 & 1 & 0 & -2 & 9 & -23 \\ 0 & 0 & 1 & -1 & 5 & -13 \end{pmatrix} \tag{1.2.21.3}$$

$$\xrightarrow{R_1 \to R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & -2 & 9 & -23 \\ 0 & 0 & 1 & -1 & 5 & -13 \end{pmatrix} = \left[\mathbf{I} \mid \mathbf{A}^{-1}\right] \tag{1.2.21.4}$$

From (1.1.35.1),

$$\mathbf{A}^{-1} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix} \tag{1.2.21.5}$$

The system is $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 11 \\ -5 \\ -3 \end{pmatrix}$. Since $\text{rank}\,(\mathbf{A}) = \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) = 3$, from (1.1.26.2) the solution is unique. From (1.1.35.1),

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix} \begin{pmatrix} 11 \\ -5 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \tag{1.2.21.6}$$

Therefore,

$$x = 1, \quad y = 2, \quad z = 3 \tag{1.2.21.7}$$

The following code verifies the solution

codes/inverse.py